

Research article

Peptidergic nature of nociception-related projections from the hypothalamic paraventricular nucleus to the dorsal horn of the spinal cord

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ABSTRACT

Hypothalamic paraventricular nucleus (PVN) projections to the spinal dorsal horn (SDH) are related to antinociception. Several neuropeptides from this nucleus could be released to the spinal cord after nociceptive stimuli. Indeed, it has been shown that enkephalins, oxytocin and vasopressin could be released at this level. Although the antinociceptive effects of these neuropeptides are well studied, little is known about the potential interaction between these molecules. In this study, we provide anatomical evidence of the interaction between oxytocin (OT), vasopressin (AVP), dynorphin (DYN) and enkephalin (ENK) along the PVN projections to the spinal dorsal horn at L3 level. A retrograde tracer (True Blue[®]) microinjected at L3 in the SDH and immunofluorescence with antibodies against OT, AVP, DYN and ENK were used. The experiments showed different levels of peptide immunoreactivity distribution along the rostro-caudal area of the PVN. A high percentage of co-localizations between two of the peptides (OT-AVP, OT-DYN, AVP-ENK, DYN-ENK) were present along the PVN. The following co-localizations occupied 4.76–9.62% of the total PVN area. PVN projections to the SDH at L3 level showed similar results. Our results show that different antinociceptive peptides may be interacting with each other to evoke PVN antinociceptive effects as part of the endogenous system of nociceptive modulation.

1. Introduction

The hypothalamic paraventricular nucleus (PVN) is a key center involved in the regulation of autonomic functions [11], satiety [19], reproduction [31] and more recently nociception [7–10,23]. PVN forms part of the anterior portion of the hypothalamus and is primarily divided into two neuronal subsets according to their size; i.e. magnocellular neurons located in the lateral part of the PVN and parvocellular neurons that occupy mainly the medial part; however, parvocellular neurons are existent all over the PVN [1,30].

Magnocellular neurons are known for their classic neuroendocrinological function mediated by oxytocin (OT) and vasopressin (AVP) secretion into the general circulation through the medial eminence and the posterior pituitary gland [2,5,25]. In addition, parvocellular neurons send projections to the dorsal horn of the spinal cord (SDH) [28], an area associated with the process of nociception. Certainly, PVN sends oxytocinergic projections to the SDH and upon electrical stimulation of PVN, it induces antinociception mediated by OT release [7,8,18]. Nevertheless, other studies also support the role of opioid peptides and

AVP in the PVN-induced analgesic effects [13,24]. In fact, AVP and OT neurons can excrete dynorphin (DYN) and enkephalin (ENK) peptides [5], molecules related with antinociceptive actions. Accordingly, intra-PVN glutamate administration increases endogenous opioid levels in the spinal cord and, consequently, increases pain thresholds [33]. Furthermore, DYN and ENK are present in the superficial layers of the SDH, where they may have a role in nociception [3,4,15,17]. In the SDH, nociceptive-specific neurons and wide dynamic range neurons, but not non-nociceptive neurons, show immunoreactivity to ENK [16].

The aim of this study was to demonstrate the localization, and co-localization, of OT, AVP, DYN and ENK neurons inside the PVN and the presence of the mentioned peptides in PVN projections to the dorsal horn of the spinal cord (SC).

2. Material and methods

2.1. Animals

A total of 26 male Wistar rats (280–300 g) from the Institute of

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¹ URL: http://132.248.142.23/web_site/home_pages/35?locale=en.

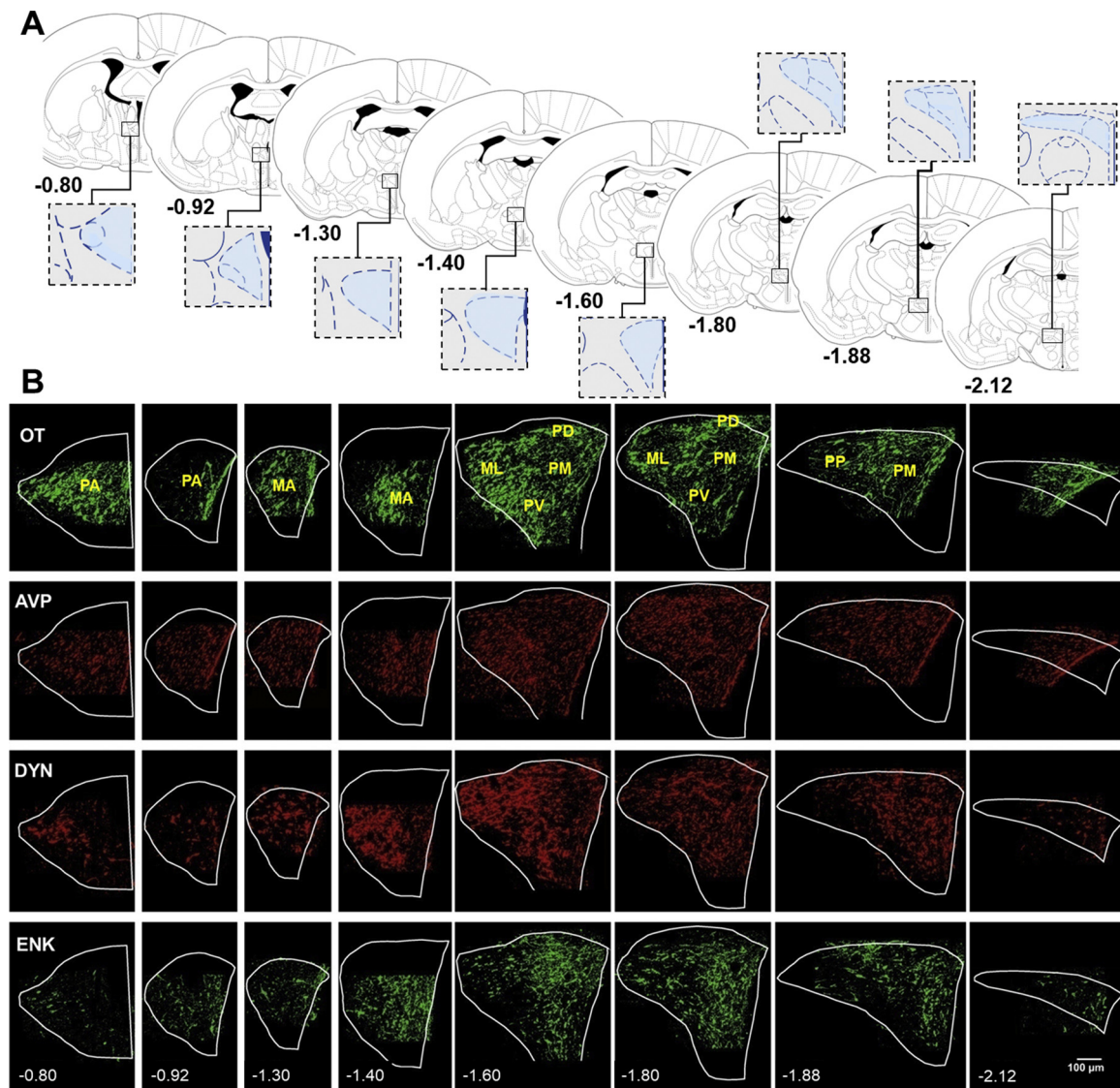


Fig. 1. (A) Images obtained from the Paxinos and Watson atlas (2007) which demonstrate the rostro-caudal extension of the PVN and the changes in its morphology along its extension. (B) Photomicrographs of OT, AVP, DYN and ENK immunoreactivity along the PVN. The outlines represent the total PVN area at each rostro-caudal level, obtained by the Cresyl-stained series in the same animal. The OT and AVP immunoreactivity images were taken from the same animal, while the DYN and ENK immunoreactivity images were taken from different animals. We can observe a higher immunoreactivity of all the neuropeptides at Bregma -1.4 , -1.6 and -1.8 levels. PA: parvocellular anterior, MA: magnocellular anterior, ML: magnocellular lateral, PM: parvocellular medial, PD: parvocellular dorsal, PV: parvocellular ventral and PP: parvocellular posterior.

Neurobiology's Animal House were used in the experiments. Animals were maintained under a 12:12 h light/dark cycle (lights on at 07:00 h) and housed in a special room at constant temperature ($22 \pm 2^\circ\text{C}$) and humidity (50%) with food and water *ad libitum*. All animal procedures and protocols were approved by our Institutional Ethics Committee, and they followed IASP ethical guidelines [35] and the Guide for the Care and Use of Laboratory Animals established by the NIH. All efforts were made to limit distress and use only the animals necessary to produce reliable scientific data.

2.2. Surgical procedures and tracer injection

Rats were anesthetized with ketamine/xylazine mixture (70/6 mg/kg, given intraperitoneal, i.p.) and then mounted onto a stereotaxic frame. Under these conditions, a partial laminectomy was performed at T12-L1 to expose the spinal cord at L3 level; the meninges were then carefully pierced. Under these conditions, True Blue® (TB) granules (retrograde tracer) were placed into superficial layers of the spinal

dorsal horn of the L3 spinal cord segment under microscopic control. The TB was deposited using a stainless-steel wire charging tiny pellets of TB attached to its blunt tip. Immediately after TB injection, an absorbable gelatin sponge (Gelfoam®; Pharmacia & Upjohn Company, MI, USA) was placed above the injection site to avoid bleeding and stitching was performed. At the end of the surgery, the animals were kept in individual cages with food and water *ad libitum*.

2.3. Perfusion, immunofluorescence, confocal microscopy and image analysis

After 12 days, the animals were re-anesthetized with ketamine/xylazine, and colchicine ($240\text{ }\mu\text{g}$ in $240\text{ }\mu\text{l}$ of purified water) was stereotactically injected into the right lateral ventricle to improve the detection of DYN and ENK neurons [12]. Twenty-four hours later, the animals were deeply anesthetized with pentobarbital (45 mg/kg ; i.p.) and perfused through the heart using a peristaltic pump with isotonic saline solution (0.9% NaCl) followed by 4% paraformaldehyde (0.1 M

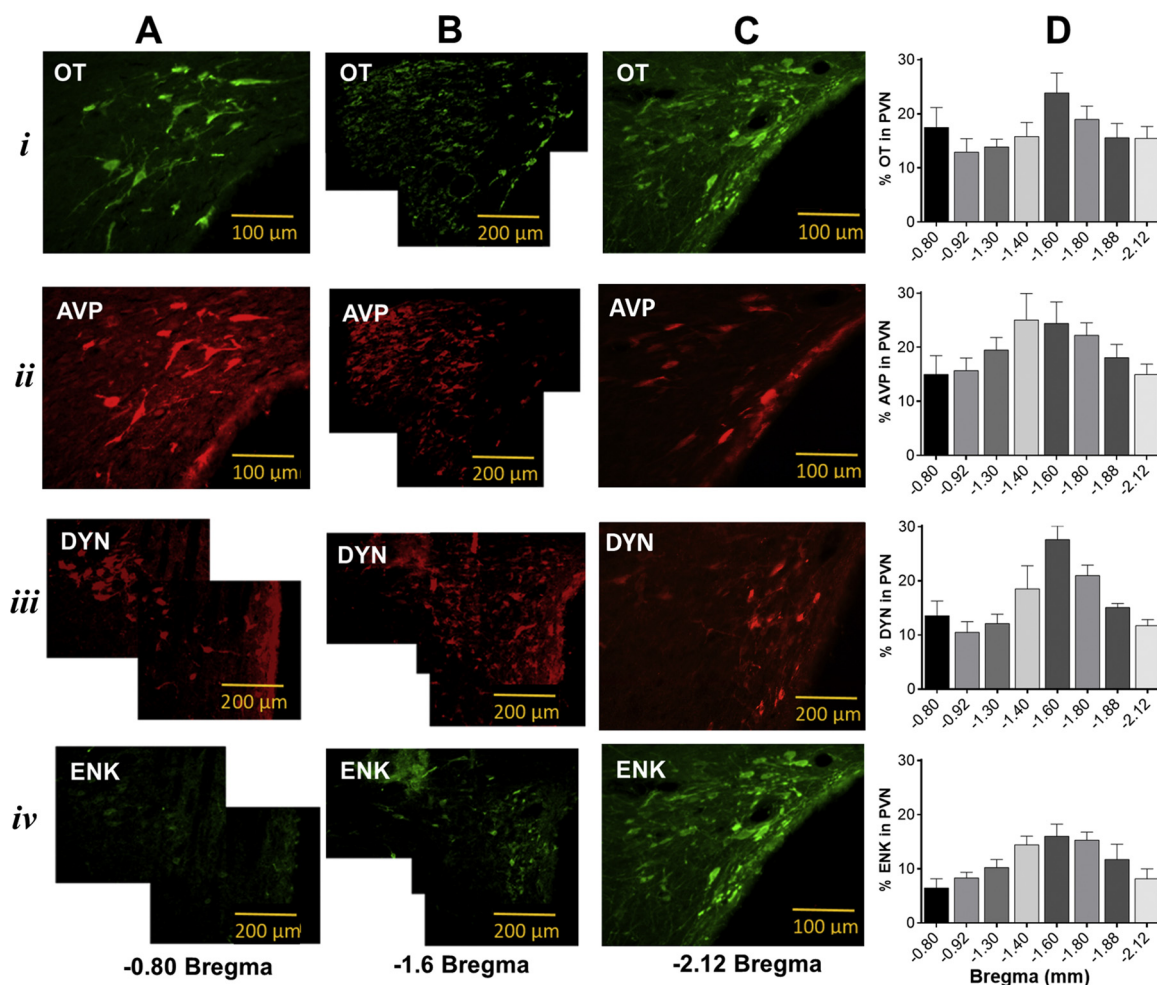


Fig. 2. Photomicrographs and distribution area (expressed in percentage) of OT, AVP, DYN and ENK immunoreactivity in the PVN. Panel (A) shows the areas of immunoreactivity for each neuropeptide at Bregma levels (mm) -0.80 (i) OT (in green), (ii) AVP (in red), (iii) DYN (in red) and (iv) ENK (in green), Panel (B) and Panel (C) show the same at Bregma -1.6 and -2.12 levels consecutively; in order to illustrate the distribution of the above neuropeptides, some panels were composed of multiple photomicrographs to cover the whole PVN area. Panel (D) demonstrates the percentage of PVN areas reactive to the mentioned peptides compared to the total area of PVN at each rostro-caudal level. Note that the levels of maximum immunoreactivity are at Bregma (mm) -1.4 , -1.6 and -1.8 levels. Furthermore, the photomicrographs are representative images taken from 1 animal ($n = 3$ rats). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

phosphate buffer, PH: 7.4). The brains and spinal cord were removed, stored in vials containing the same fixative for an additional 2 h, and cryoprotected in a 30% sucrose buffer solution.

Subsequently, the brain (Bregma -0.80 to -2.30 , according to Paxinos and Watson [20], and spinal cord were frozen and cut into 40 μ m-thick coronal sections with a freezing microtome (Leica SM 2000 R; Leica Instruments BmbH, Nussloch, Germany). The slides were collected in 5 series, the first one to observe the site of injection and the marked cells in PVN. The other 4 groups were used to perform immunofluorescence for OT, AVP, DYN and ENK along the PVN.

Eight coronal sections of PVN of each series were pre-incubated for 20 min in goat serum in phosphate buffered saline (PBS) with triton (5%) and incubated overnight at 4°C using the following primary antibodies: (1) Anti OT-neurophysin (PS60, mouse, monoclonal, diluted 1:250; a gift from Dr. H. Gainer from the National Institute of Neurological Disorders and Stroke, NIH, Bethesda MD); (2) Anti-AVP (polyclonal, rabbit, diluted 1:500; Cat.no. AB1565, Millipore, Chemicon International, Temecula CA); (3) Anti-ENK (monoclonal, mouse, diluted 1:1000; Cat.no. MAB350, Millipore, Chemicon International, Temecula, CA); (4) Anti-DYN (polyclonal, rabbit, diluted 1:200; Cat no. T-4268, Bachem Group, Peninsula Laboratories, CA). The combinations for double labeling were OT-AVP, OT-DYN, AVP-ENK, and DYN-ENK,

whereas the OT-ENK and AVP-DYN combinations were not performed due to the same animal origin of the antibodies. Hence, the sections were washed for 5 min with 0.1 M PBS and then incubated with the appropriate secondary antibody for 24 h at 4°C . The secondary antibodies used were Alexa Fluor[®] 555 goat anti-rabbit, diluted 1:300 (Cat. No. A21428, Molecular Probes, Life Technologies, Oregon, USA) and Alexa Fluor[®] 488 goat anti-mouse, diluted 1:300 (Cat. No. A11001, Molecular Probes, Life Technologies, Oregon, USA) depending on the experimental conditions.

2.4. Confocal microscopy and image analysis

Confocal images of the slides were acquired, processed and observed using the inverted LSM 510 Meta confocal microscope (Zeiss, Mexico) with 25X objective. The 488-nm argon laser was used to excite the Alexa Fluor[®] 488 signal and Laser DPSS- 561 was used to excite Alexa Fluor[®] 555. Optical sections were acquired at a digital size of 1024×1024 pixels and averaged three times to reduce noise. In all cases, the image obtained was improved (brightness and gamma). The color for each channel was selected as follows: green for OT or ENK and red for AVP or DYN. In all cases, a two-dimensional projection image was imported in Tag Image File Format (TIFF) and were used to

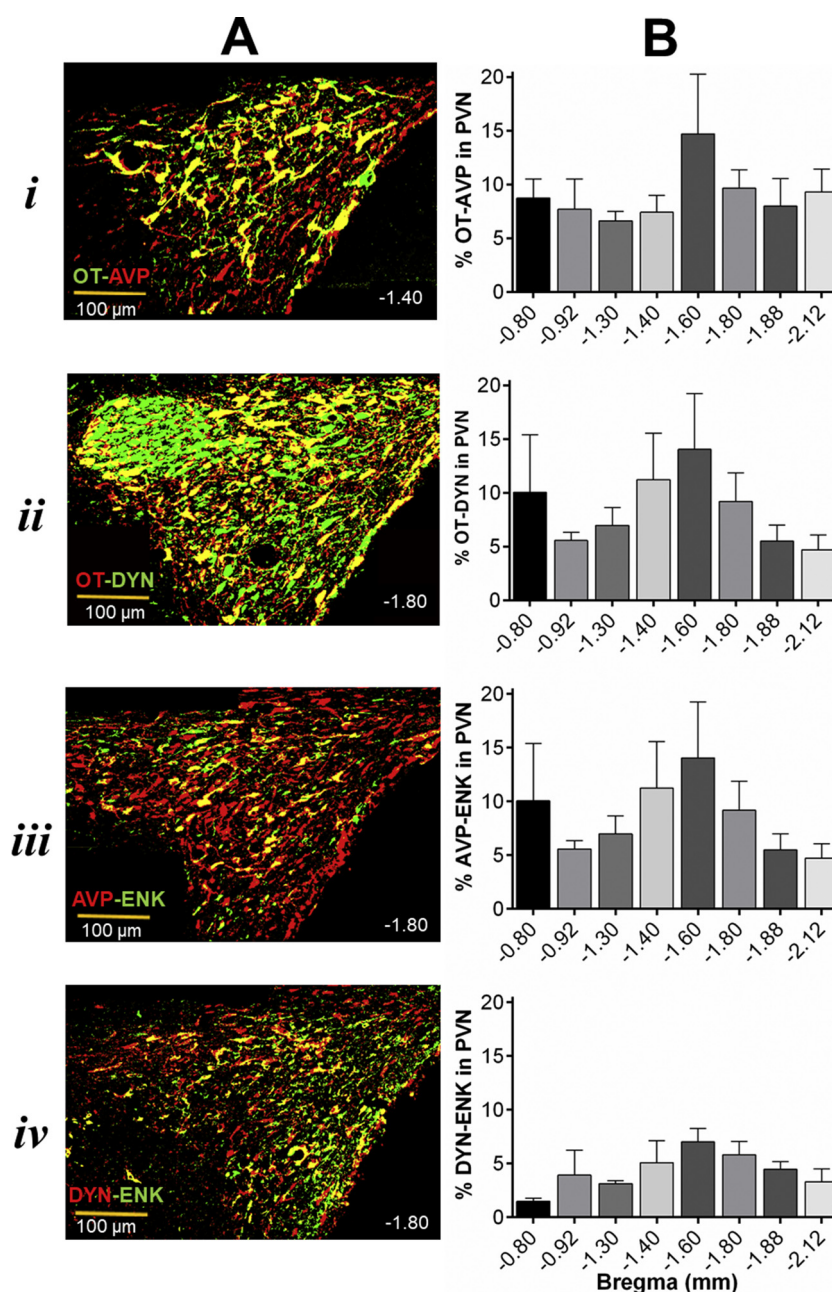


Fig. 3. Photomicrographs and distribution area (expressed in percentage) of double colocalization of the neuropeptides. (A) The areas in yellow show the regions of the PVN with double labeling immunofluorescence. Panel (B) demonstrates the percentage of each co-localization area (OT-AVP, OT-DYN, AVP-ENK and DYN-ENK) compared to the total PVN area at different rostro-caudal levels. Furthermore, the photomicrographs are representative images taken from 1 animal ($n = 3$ rats). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

compose the multi-panel figures. Furthermore, we obtained photos using the Leica DFC320 digital camera system. The amounts of positive cells were compared for the different peptides and also the area of peptides fluorescence were compared with the total area of the PVN.

2.5. Statistical analysis

The immunoreactivity and co-localizations were quantified using ImageJ software (V 1.44o, Java 1.6.0.20 (32-bit), NIH, USA) [26]. The data are presented as mean $\pm$ standard error of the mean (SEM) and, considering that the data obtained were non-parametric, we used the Kruskal-Wallis test to determine statistical differences between two or more Bregma areas (if necessary) followed by Dunn's post hoc test. Statistical significance was accepted at $p < 0.05$.

3. Results

3.1. Neuropeptide distribution along the PVN

As shown in Figs. 1 and 2, the immunoreactivity distribution analysis for OT, AVP, ENK and DYN along the PVN indicates that all the mentioned peptides show a preferential immunoreactivity at Bregma -1.60 level ($n = 3$). The quantification of immunoreactivity areas compared to the total PVN area (Bregma -0.80 to -2.30) demonstrates that 17.08% of the PVN is immunoreactive to OT (OT+ve), while 20.39% is AVP+ve, 18.21% is DYN+ve and 12.43% is ENK+ve.

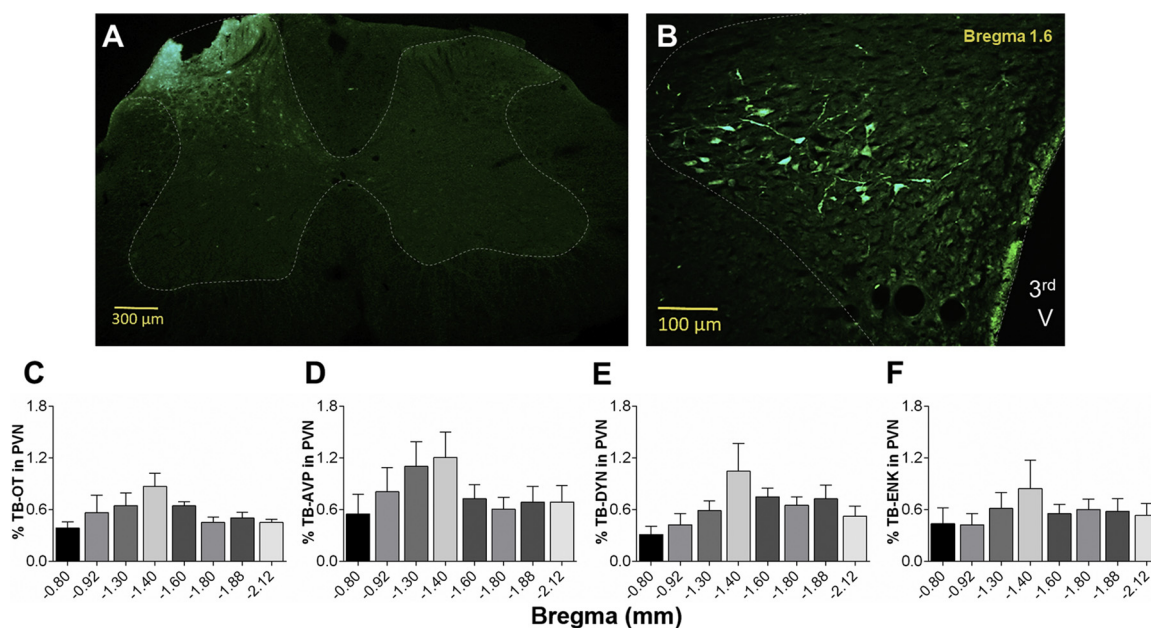


Fig. 4. (A) Photomicrograph showing the site of injection of True Blue (TB) retrograde tracer into the superficial layers of the dorsal horn of the spinal cord at L3 region. Panel (B) demonstrates TB +ve cells in the PVN. Panel (C) shows the percentage of co-localization of TB +ve cells with OT +ve cells (TB-OT), (D) shows (TB-AVP) co-localization, (E) (TB-DYN) co-localization and (F) (TB-ENK) co-localization, each compared to the total PVN area at different rostro-caudal levels. Furthermore, the photomicrographs are representative images taken from 1 animal (n = 3 rats).

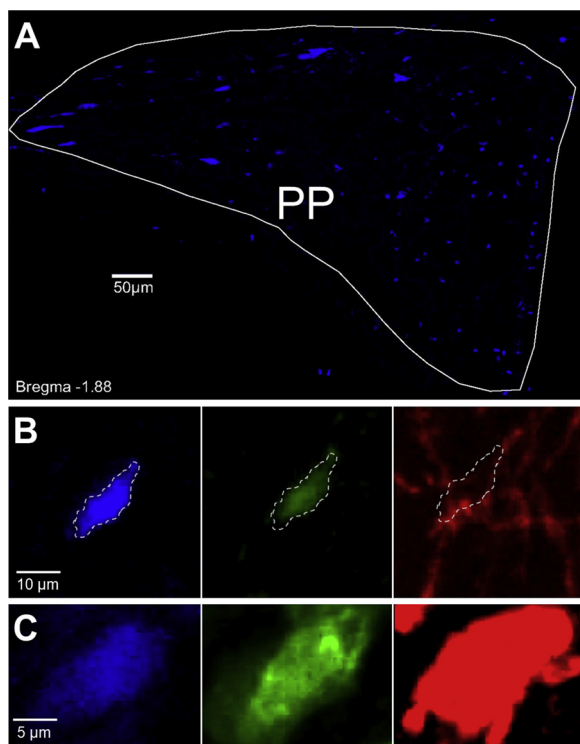


Fig. 5. PVN projecting neurons to the spinal cord. (A) in blue color, we observe TB +ve cells (PVN neurons that project to the SDH at L3 level). This is a reconstructed image from photomicrographs taken by a fluorescence microscope at 20 $\times$ object. (B) an example of TB-OT co-localization, in blue a PVN projecting cell to the spinal cord (TB +ve), the same cell in green is OT +ve and in the right corner the phantom of the same cell is not positive to AVP (AVP+ve) (Bregma -1.6). (C) a TB-DYN-OT co-localization example, at the left a PVN projecting cell in blue (TB +ve), at the middle the same cell in green is DYN +ve and at the right side the same cell in red is OT +ve (Bregma -1.3). (B) and (C) images were obtained from the same animal. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

3.2. Neuropeptides co-localization within the PVN

Fig. 3 shows the neuropeptides co-localization along the PVN. It is interesting to note that OT-AVP (Fig. 3Ai), OT-DYN (Fig. 3Aii), AVP-ENK (Fig. 3Aiii) and DYN-ENK (Fig. 3Aiv) are preferentially co-localized at Bregma -1.6 and/or -1.4. When we analyzed the global co-localization, we found that OT-AVP was present in 9.62% of the total PVN area, but its highest co-localization was observed at Bregma -1.60 level. OT-DYN co-localization occupied 9.23% of the PVN associated with high reactivity in the posterior portion of the PVN. In contrast, AVP-ENK co-localization marks 5.79% whereas DYN-ENK co-localization is observed in only 4.76% of the total PVN area. It is important to point out that: (i) OT-DYN or AVP-ENK co-localize along the parvocellular anterior (PA), magnocellular lateral (ML) and parvocellular medial (PM) portions of the PVN; whereas (ii) OT-AVP or DYN-ENK, co-localize throughout the PVN, especially in the PM region (data not shown).

3.3. Site of TB injection and the neuropeptide distribution of PVN projection to the SDH

PVN projections to the SDH were assessed using TB injections at L3. As shown in Fig. 4A, TB granules were injected into the superficial layers of the SDH at L3 level. It is interesting to note that after 13 days, the TB retrograde tracer was successfully carried to PVN (Fig. 4B). The TB +ve cells appear along the PVN and increase in the PM portion; furthermore, they are also expressed, to a lesser extent, in the parvocellular posterior (PP) area (see Fig. 5).

Globally, TB injected at SDH and transported to PVN only occupy 5.06% of the total PVN area. Indeed, inferior panels of the Fig. 4 shows the colocalization between TB and OT (Fig. 4C), AVP (Fig. 4D) DYN (Fig. 4E) or ENK (Fig. 4F) in the PVN region. In all cases, the maximal co-localization evaluated as area of stain was found between Bregma -1.40 and -1.80. When we analyzed the global peptidergic nature of TB +ve cells area in the PVN we found that 11.06% of the TB +ve area was also OT +ve (TB-OT), 15.01% was AVP +ve (TB-AVP), 11.46% was DYN +ve (TB-DYN) and 13.04% was ENK +ve (TB-ENK).

Finally, Fig. 6 show the double neuropeptide co-localization along

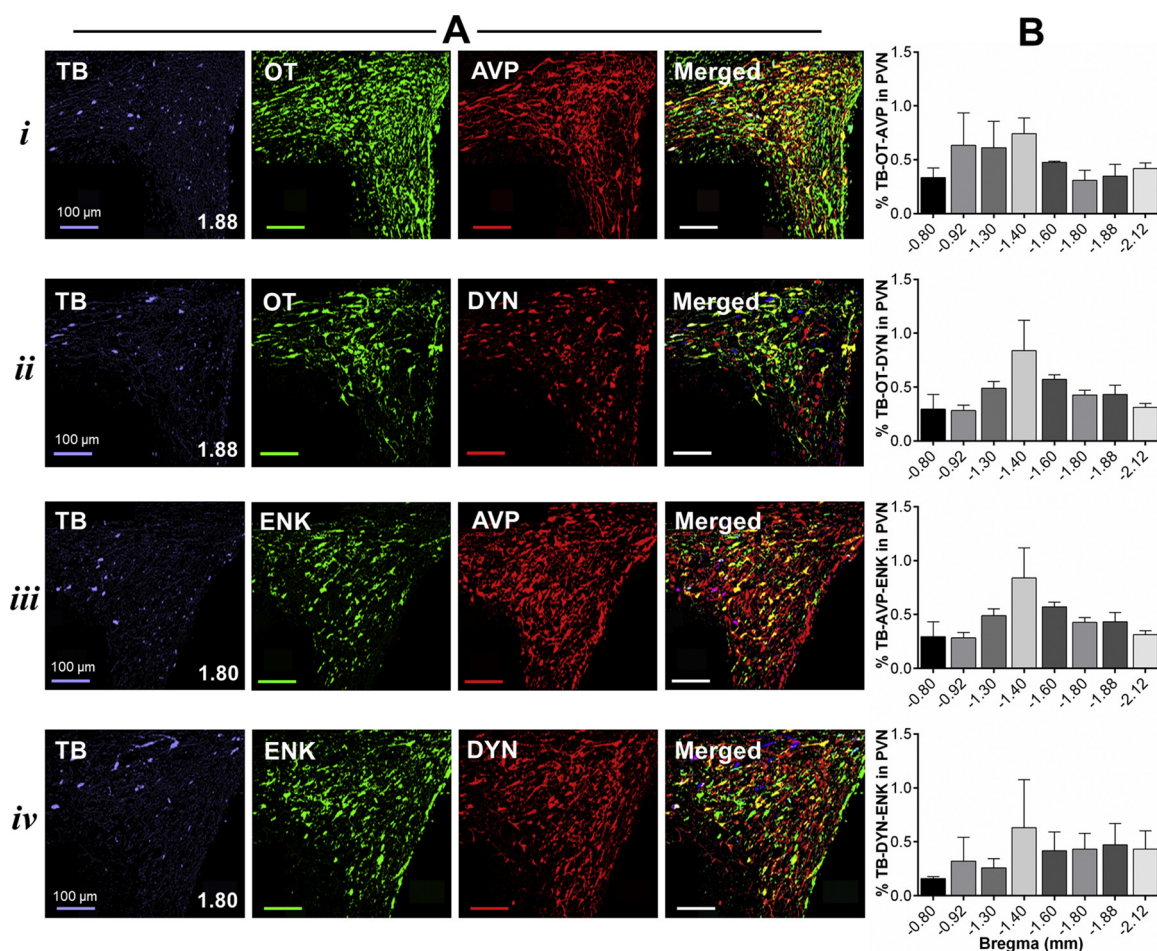


Fig. 6. Panel (A) shows photomicrographs with triple labeling immunofluorescence. Panels in (B) demonstrate the percentage of each co-localization compared to the total PVN area at different rostro-caudal levels. Furthermore, the photomicrographs are representative images taken from 1 animal ($n = 3$ rats).

the PVN. Globally, 9.10% of the TB+ve area was reactive to both OT and AVP (TB-OT-AVP), 9.52% was OT and DYN+ve (TB-OT-DYN), 7.64% was AVP and ENK+ve (TB-AVP-ENK) and 8.10% was DYN and ENK+ve (TB-DYN-ENK). It is important to notice, in Fig. 6, that TB+ve cells bodies are located in the VPN and they send projections to the SDH.

4. Discussion

The PVN projects to many areas in the central nervous system; for instance, the medial eminence, ventral tegmental area, Edinger-Westphal nucleus, periaqueductal grey (PAG) matter, superior salivary nucleus, medial solitary tract nucleus, amygdala and SC [14,22,27]. Numerous neuropeptides participate in these projections. In the current study, we focused on nociception-related peptides (OT, AVP, ENK and DYN) and their co-localization along the PVN. The following co-localizations occupy 4.76–9.62% of the total PVN area with no significant differences. We believe that this could be a possible mechanism to regulate peptide secretion within the PVN. In this context, DYN is believed to regulate AVP secretion in magnocellular neurons of the supraoptic nucleus (SON) [6]. OT and AVP can also regulate opioid peptides, while OT and AVP administration into the PAG increases leu-ENK and met-ENK in the same region [32,34]; thus, the double peptide expression of PVN neurons could be related to the regulation of secretion and other associated functions.

Furthermore, PVN projections reach superficial laminae of the dorsal horn, the intermediate column and the central canal of the SC [28]. The density of these projections to the superficial layers of the

dorsal horn increases at cervical segment level (104 neurons) [7] and decreases as they reach the caudal region (13 neurons at L1 level and 57 neurons at L4 and L5 levels [7]. These PVN projections to the superficial layers of the SC dorsal horn are important in the modulation of nociception, as electrical stimulation of the PVN regenerates an antinociceptive response in the presence of nociceptive stimuli [8,9]. OT fibers participate in the process, as the antinociceptive effect was attenuated by the OTR blockade. Rojas-Piloni et al. [23] pointed out that this nociceptive modulation is carried out by OT but not by AVP, and is thought to be mediated by GABA interneurons. Additionally, the authors describe that the presence of OT+ve fibers along SC segments is greater than that of AVP+ve fibers, and that intrathecal administration of AVP did not produce analgesia [23]. However, other studies suggest a possible role of AVP in nociception [21,29]. The electrophysiological results could not define the role of AVP in this process.

In this study, we demonstrate the presence of other neuropeptides that take part in the PVN-SC connection at L3 level. The existence of AVP+ve fibers could be related to the somatosensory process or to nociception. DYN+ve and ENK+ve fibers support a possible role of opioid peptides in the analgesia induced by PVN stimulation. Intrathecal administration of Naloxone reduced the analgesic effect induced by electrical stimulation of the PVN [18].

Our study indicates the percentage of OT, AVP, DYN and ENK immune-reactive areas in the PVN and the amount of neuropeptide fibers they share in the PVN-SC projections. We also demonstrate the co-localization of the peptides (OT-AVP, OT-DYN, AVP-ENK and DYN-ENK) along the PVN. Since we did not perform the co-localization of the four peptides together in the same animal, we cannot signal the exact

expression area of each one. Nevertheless, we consider that the high level of co-localization between the mentioned peptides shows that PVN neurons could release two peptides together to perform a certain function (e.g. nociceptive modulation). We also think that the distribution of the peptides in the dorsal horn of the SC could be related to other somatosensory functions in addition to the modulation of nociception, and that these functions might be regulated by at least two peptides. Taken together, the presence of various neuropeptides arising from the same nucleus (e.g., PVN) or from different nuclei at spinal level emphasizes the importance of their interactions and the redundancy of the endogenous system of nociceptive modulation.

5. Conclusion

The results show extended co-localization between OT, AVP, DYN and ENK peptides in the PVN as well as in PVN projections to the dorsal horn of the SC at L3 level. A portion (11.06%) of PVN projections into the SC is oxytocinergic; this represents 0.56% of the total PVN area. Although this number may seem low, the projections are responsible for pain modulation at spinal level, produce antinociceptive effects and increasing pain thresholds. Moreover, we demonstrate that other peptides cannot be excluded. Finally, OT, AVP, DYN and ENK projections from the PVN and other brain nuclei into the dorsal horn of the SC could be important for the nociception process, in which they might interact to ensure an effective and potent effect from the endogenous system of nociceptive modulation.

Conflict of interest

The authors declare that they have no conflict of interest.

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